Lesson 2 — Statistical Learning: Parameter Estimation

MLE, Method of Moments, Fisher Information, Uncertainty

Applied Statistics Course

October 15, 2025

Learning Objectives

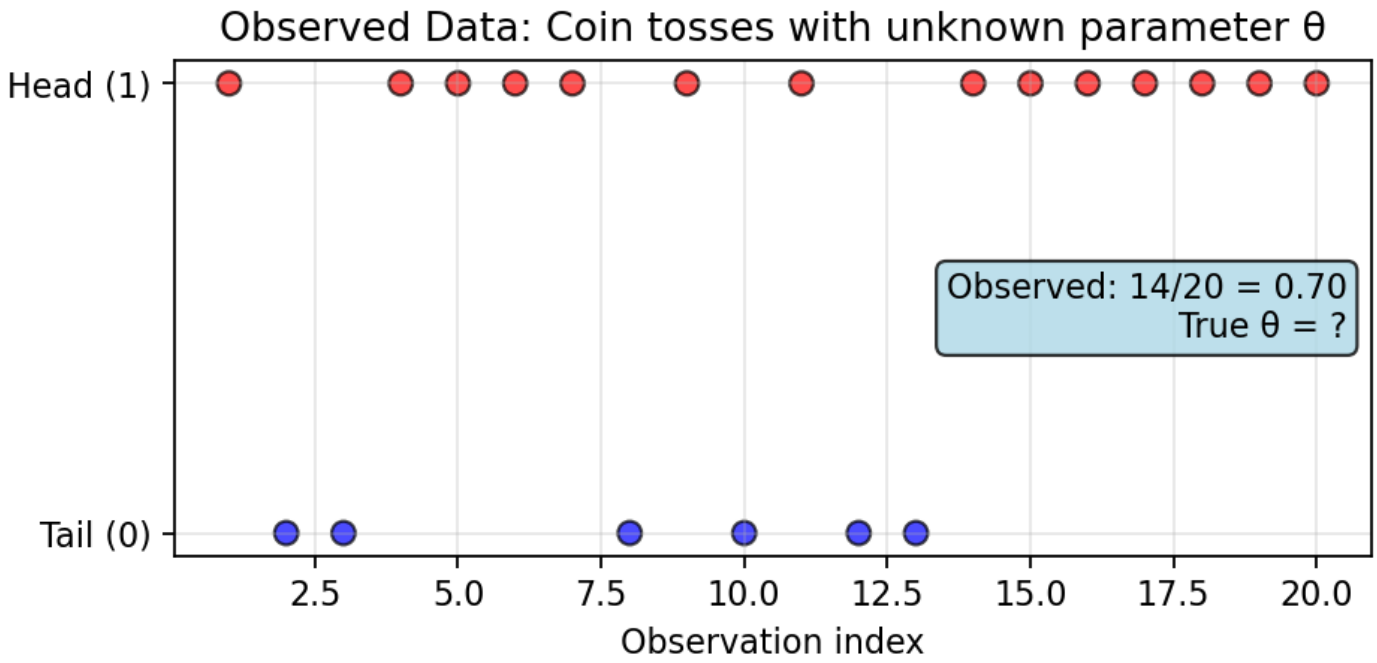
- Derive estimators using **maximum likelihood** and **method of moments**.
- Compute **standard errors** via Fisher information and the **delta method**.
- Assess estimators: **bias**, **variance**, **MSE**; use **likelihood profiles** and **bootstrap**.
- Implement simulation studies to compare procedures.

Recall Lesson 1: random variables, laws (PMF/PDF/CDF), LLN/CLT, and notation ($\mathbb{P}, \mathbb{E}, \mathrm{Var}$).

Outline

1. Maximum Likelihood Estimation (MLE)
   - Likelihood
   - Maximum Likelihood Estimate
2. Method of Moments

Parameter Estimation: Why Do We Care?

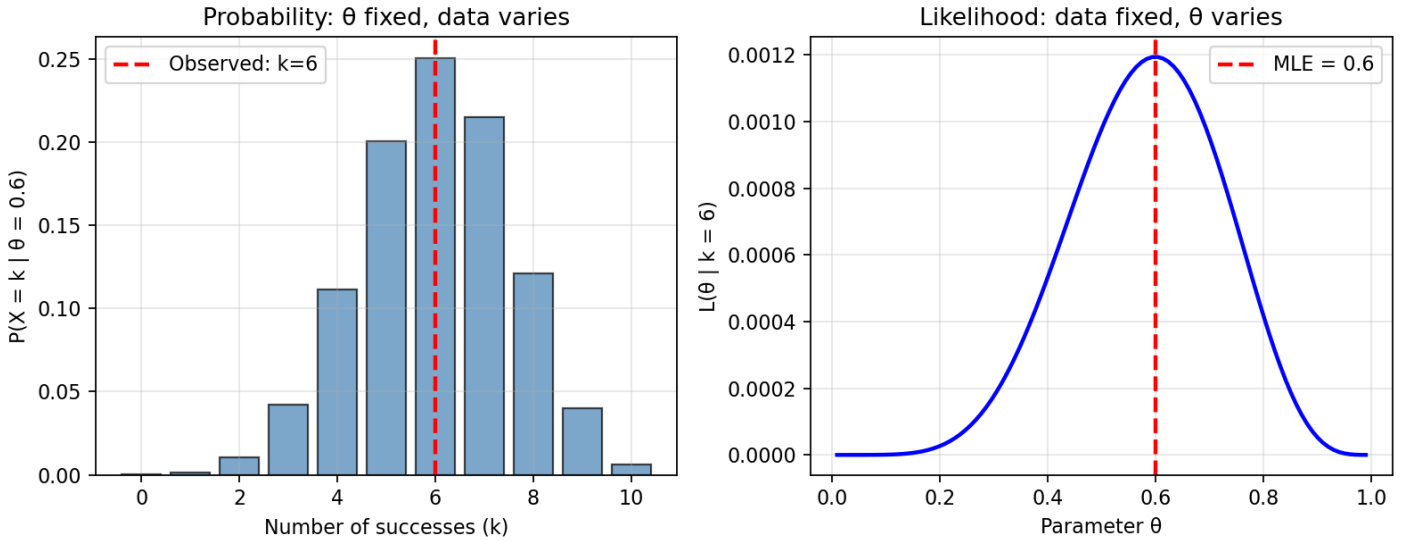


The Central Question

Given observed data, what parameter values make this data most plausible under our model?

- We have data $x_1, x_2, \ldots, x_n$
- We assume a probabilistic model $f(x \mid \theta)$
- We want to find the "best" estimate $\hat{\theta}$

Probability vs. Likelihood: The Key Duality



Probability

Fixed θ, varying data x

$P(X = x \mid \theta)$

Likelihood

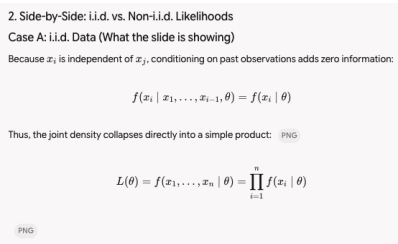
Fixed data x, varying θ

$L(\theta \mid x) \propto P(X = x \mid \theta)$

Key Insight

Same mathematical function, but we're asking different questions!

What is the Likelihood Function?


2. Side-by-Side: i.i.d. vs. Non-i.i.d. Likelihoods

Case A: i.i.d. Data (What the slide is showing)

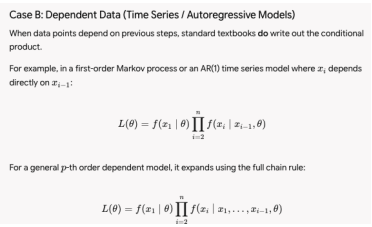
Because x_i is independent of x_j, conditioning on past observations adds zero information:

$$f(x_i \mid x_1, \ldots, x_{i-1}, \theta) = f(x_i \mid \theta)$$

Thus, the joint density collapses directly into a simple product: PNG

$$L(\theta) = f(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

PNG


Case B: Dependent Data (Time Series / Autoregressive Models)

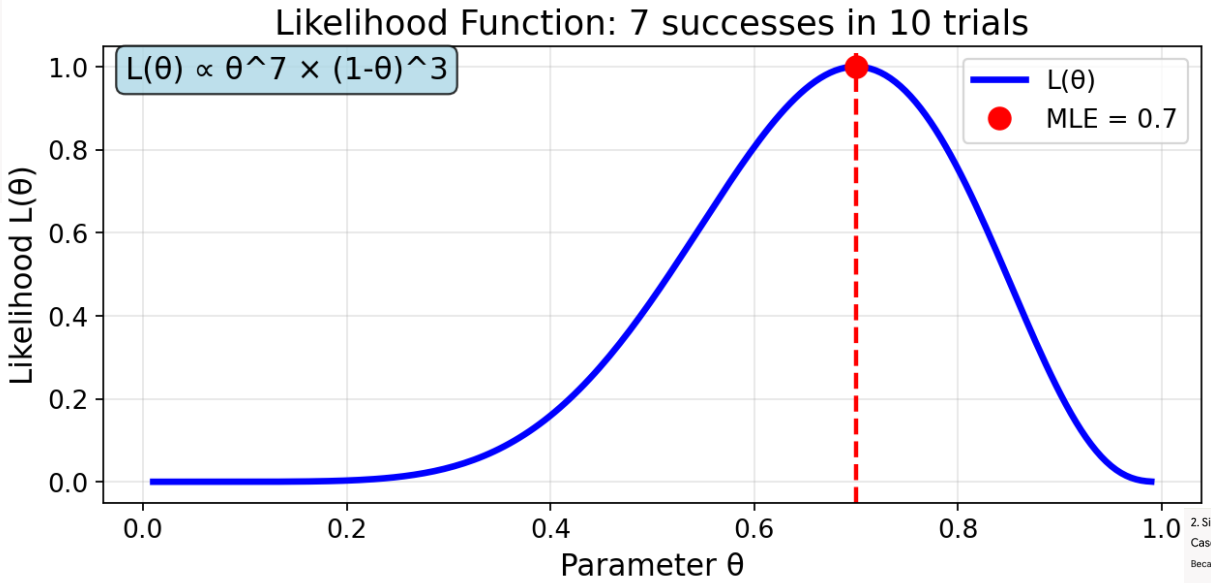
When data points depend on previous steps, standard textbooks **do** write out the conditional product.

For example, in a first-order Markov process or an AR(1) time series model where x_i depends directly on x_{i-1}:

$$L(\theta) = f(x_1 \mid \theta) \prod_{i=2}^{n} f(x_i \mid x_{i-1}, \theta)$$

For a general p-th order dependent model, it expands using the full chain rule:

$$L(\theta) = f(x_1 \mid \theta) \prod_{i=2}^{n} f(x_i \mid x_1, \ldots, x_{i-1}, \theta)$$

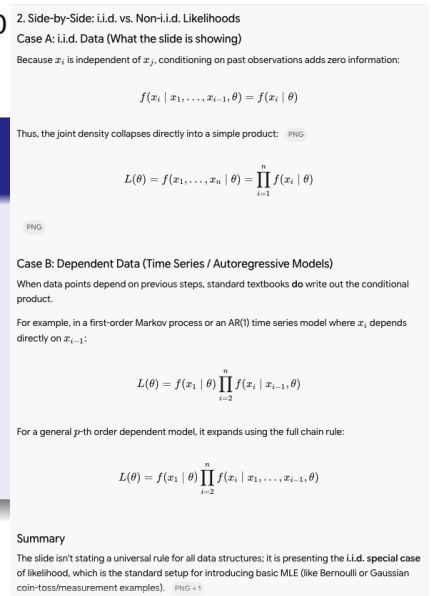


Definition

For observed data $x_1, \ldots, x_n$ and model $f(x \mid \theta)$:

$$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

- Measures how well parameter θ explains the observed data
- Higher likelihood $\Rightarrow$ parameter is more supported by data
- Often work with log-likelihood: $\ell(\theta) = \sum_{i=1}^{n} \log f(x_i \mid \theta)$


2. Side-by-Side: i.i.d. vs. Non-i.i.d. Likelihoods

Case A: i.i.d. Data (What the slide is showing)

Because x_i is independent of x_j, conditioning on past observations adds zero information:

$$f(x_i \mid x_1, \ldots, x_{i-1}, \theta) = f(x_i \mid \theta)$$

Thus, the joint density collapses directly into a simple product: PNG

$$L(\theta) = f(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

PNG

Case B: Dependent Data (Time Series / Autoregressive Models)

When data points depend on previous steps, standard textbooks **do** write out the conditional product.

For example, in a first-order Markov process or an AR(1) time series model where x_i depends directly on x_{i-1}:

$$L(\theta) = f(x_1 \mid \theta) \prod_{i=2}^{n} f(x_i \mid x_{i-1}, \theta)$$

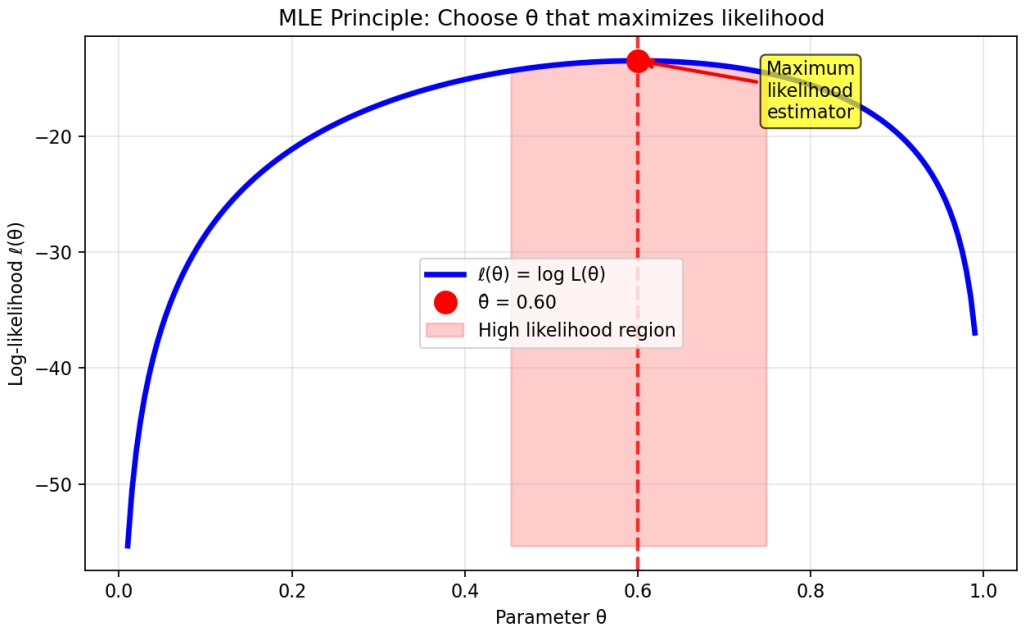
For a general p-th order dependent model, it expands using the full chain rule:

$$L(\theta) = f(x_1 \mid \theta) \prod_{i=2}^{n} f(x_i \mid x_1, \ldots, x_{i-1}, \theta)$$

Summary

The slide isn't stating a universal rule for all data structures; it is presenting the **i.i.d. special case** of likelihood, which is the standard setup for introducing basic MLE (like Bernoulli or Gaussian coin-toss/measurement examples). PNG +1

The Maximum Likelihood Principle



MLE Definition

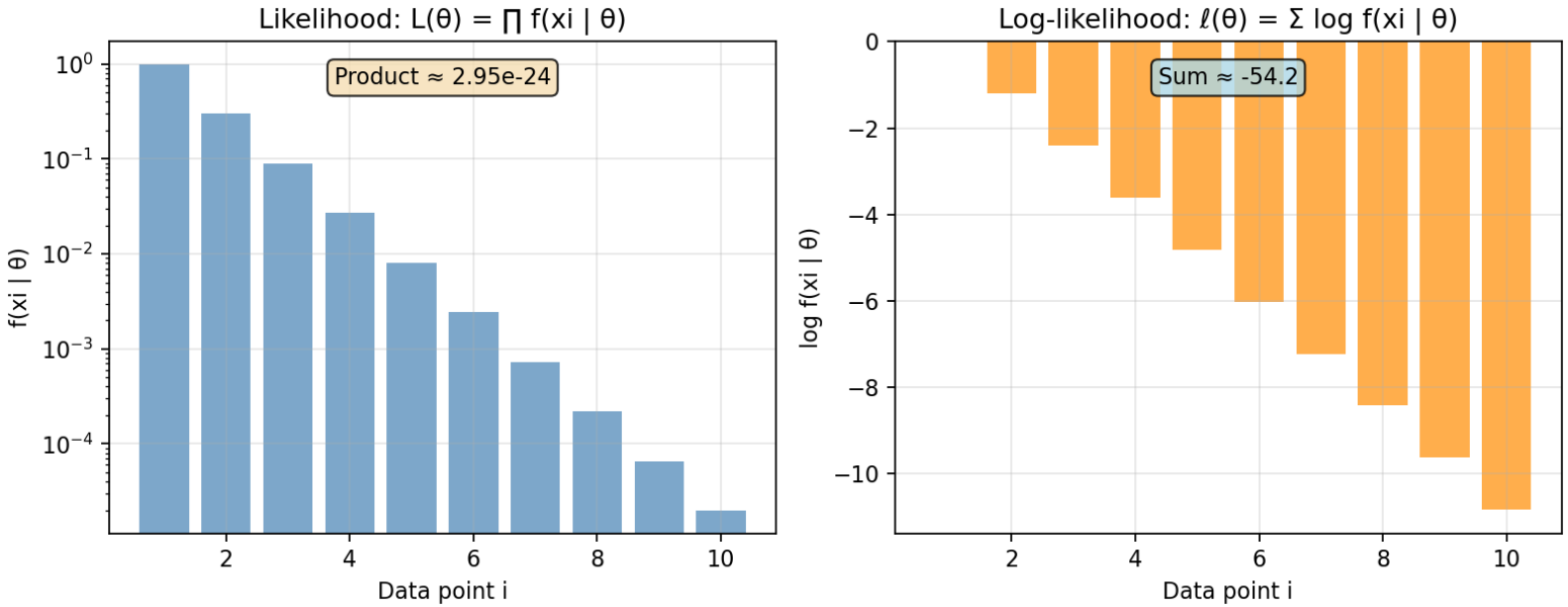
The Maximum Likelihood Estimator (MLE) is:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} L(\theta) = \arg\max_{\theta} \ell(\theta)$$

Intuition

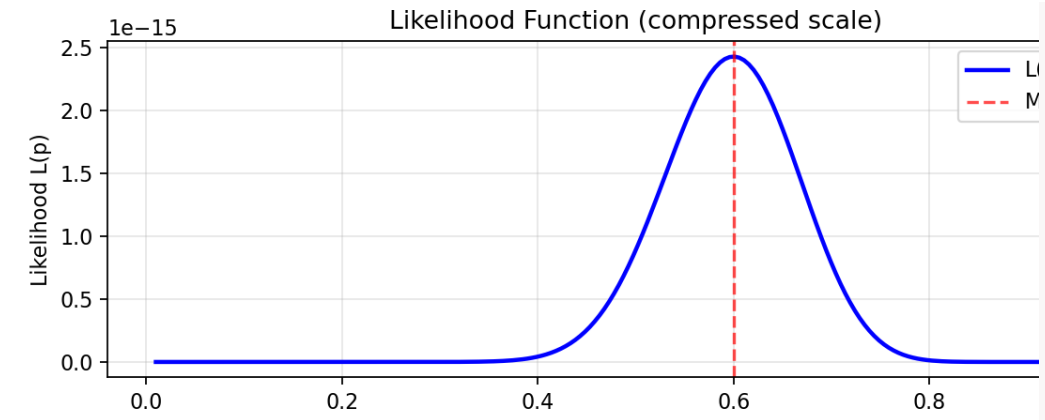
Choose the parameter value that makes our observed data as likely as possible.

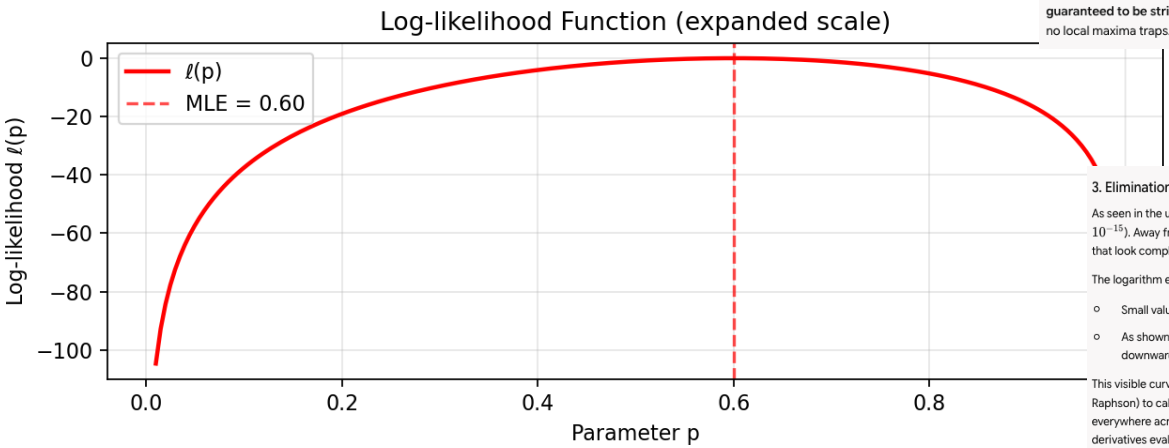
From Likelihood to Log-Likelihood



- Likelihood is a product of terms: $L(\theta) = \prod_{i=1}^{n} f_\theta(x_i)$
- Products are difficult to optimize when n is large
- Take the log to simplify: $\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f_\theta(x_i)$
- Since log is strictly increasing, maximizing L or ℓ is equivalent

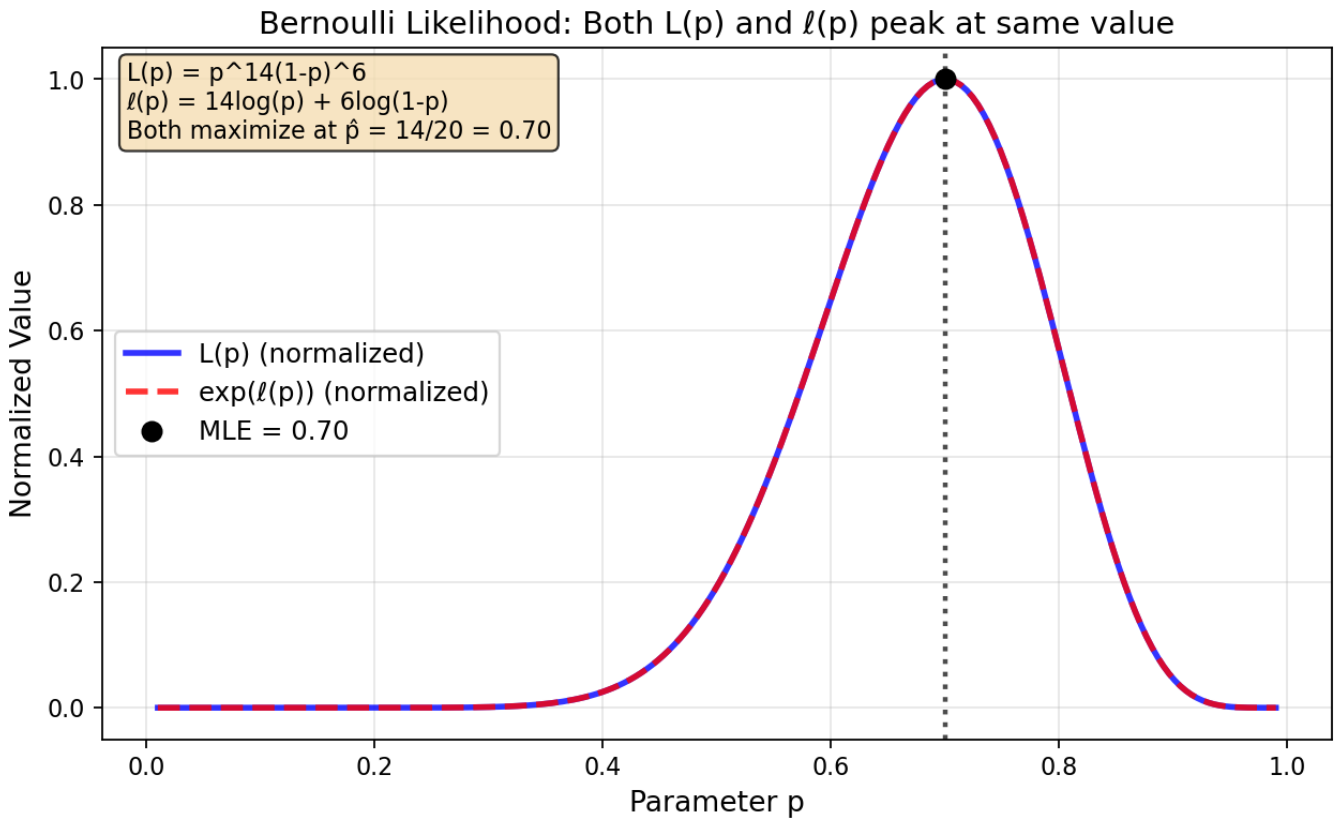
Why use log-likelihood in practice?





- **Numerical stability:** avoids underflow when multiplying many small numbers
- **Simplifies optimization:** turns products into sums, easier differentiation
- **Reveals structure:** concavity/convexity properties often clearer
- **Same maximum:** $\arg\max L(\theta) = \arg\max \ell(\theta)$

Log-Likelihood for the Bernoulli Model



Bernoulli Case

Likelihood: $L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i}$

Log-likelihood: $\ell(p) = (\sum x_i)\log p + (n - \sum x_i)\log(1-p)$

Easier to differentiate, leads to closed-form solution: $\hat{p} = \frac{1}{n}\sum x_i$

Example: Coin Flips (Bernoulli Model)

Setup: n coin flips, k heads observed. Model: $X_i \sim \text{Bernoulli}(p)$

Likelihood: $L(p) = p^k(1-p)^{n-k}$

Log-likelihood: $\ell(p) = k \log p + (n-k)\log(1-p)$

Find MLE: $\frac{d\ell}{dp} = \frac{k}{p} - \frac{n-k}{1-p} = 0$

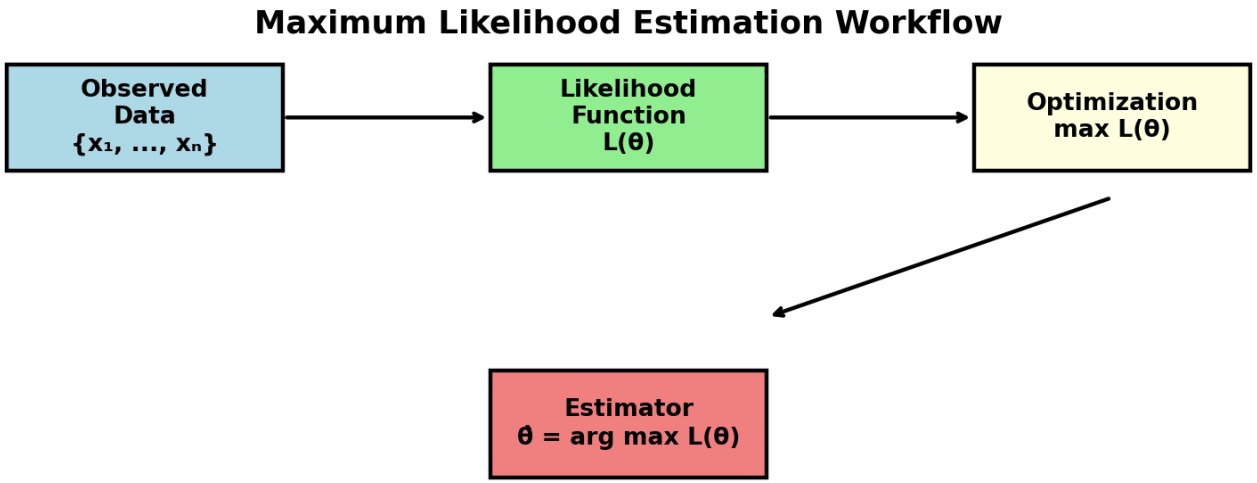
Result

$$\hat{p}_{\text{MLE}} = \frac{k}{n} = \bar{x}$$

Makes Sense!

The proportion of heads in our sample is the most likely value for the coin's bias.

MLE: Key Takeaways

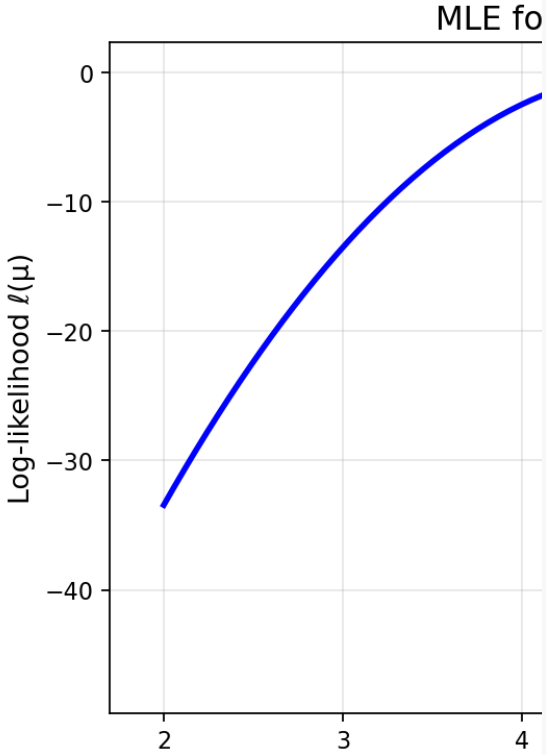


- **Intuitive:** Choose parameters that best explain observed data
- **General:** Works for any probabilistic model
- **Principled:** Solid theoretical foundation
- **Practical:** Often gives closed-form solutions

Coming Up

More examples, properties, and when MLE works well (or doesn't!)

MLE for the Normal Mean (σ^2



1. Mathematical Derivation

Given n independent and identically distributed samples $x_1, x_2, \ldots, x_n \sim \mathcal{N}(\mu, \sigma^2)$: PNG

- **Probability Density Function (PDF):**

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

- **Log-Likelihood $\ell(\mu)$:** Taking the natural logarithm of the joint distribution $\prod_{i=1}^{n} f(x_i)$ converts the product into a sum:

PNG

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

PNG

- **Maximization:** Differentiate $\ell(\mu)$ with respect to μ and set the derivative to zero:

PNG

$$\frac{d\ell(\mu)}{d\mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0$$

$$\sum_{i=1}^{n} x_i - n\mu = 0 \implies \hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}_n$$

Model & Solution

Model: $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with σ^2 known

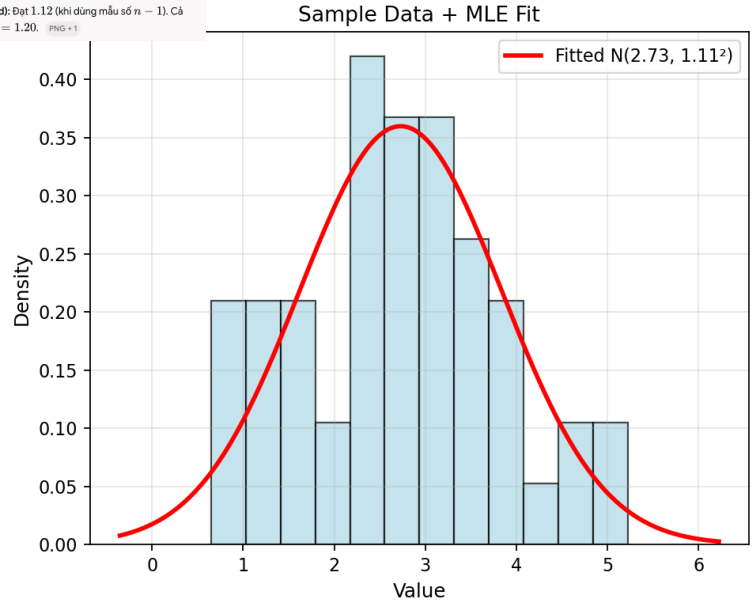
Log-likelihood: $\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$

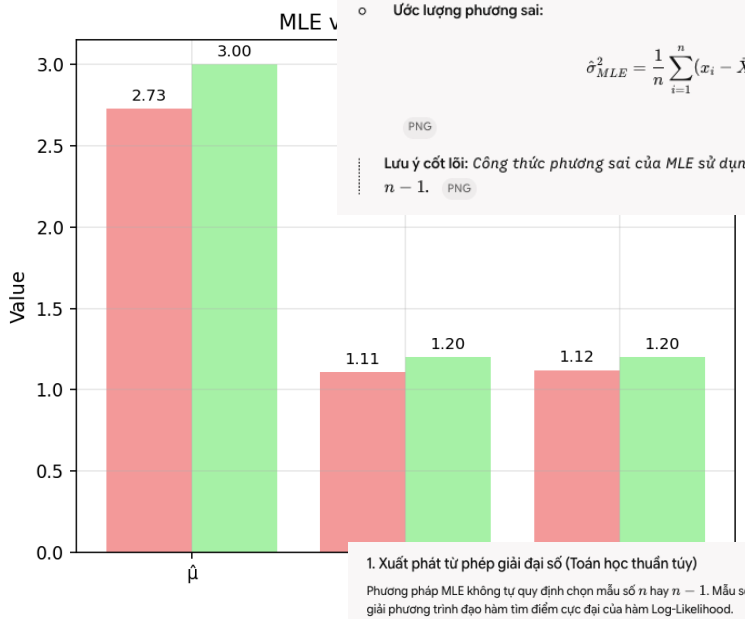
MLE: $\hat{\mu}_{MLE} = \overline{X}_n$

MLE for Normal (μ, σ^2 unknown)

2. Giải thích Chi tiết Hai Biểu đồ

- **Biểu đồ bên trái (Sample Data + MLE Fit):**
  Hiển thị biểu đồ tần suất (histogram) của tập dữ liệu mẫu và đường cong phân bố Chuẩn màu đỏ $\mathcal{N}(2.73, 1.11^2)$ được khớp tương ứng với các giá trị vừa tính toán từ MLE. PNG
- **Biểu đồ bên phải (MLE vs True Parameters):**
  So sánh trực quan giữa giá trị ước lượng từ dữ liệu (cột đỏ) và giá trị thực của quần thể (cột xanh lá): PNG
  - **Trung bình $\hat{\mu}$:** Ước lượng đạt 2.73 so với giá trị thực $\mu = 3.00$. PNG
  - **Độ lệch chuẩn MLE ($\hat{\sigma}$ MLE):** Đạt 1.11 (khi dùng mẫu số n). PNG
  - **Độ lệch chuẩn không chệch ($\hat{\sigma}$ Unbiased):** Đạt 1.12 (khi dùng mẫu số $n-1$). Cả hai đều ước lượng tiệm cận giá trị thực $\sigma = 1.20$. PNG + 1





1. Mô hình Toán học và Công thức Giải (Joint Estimation)

Cho n mẫu i.i.d. $X_1, X_2, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. Khi lấy đạo hàm riêng của hàm Log-Likelihood $\ell(\mu, \sigma^2)$ theo lần lượt μ và σ^2 rồi cho bằng 0, ta thu được nghiệm dạng đóng: PNG + 1

- **Ước lượng trung bình:**

$$\hat{\mu}_{MLE} = \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} x_i$$

PNG

- **Ước lượng phương sai:**

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{X}_n)^2$$

PNG

> **Lưu ý cốt lõi:** *Công thức phương sai của MLE sử dụng mẫu số là n* **chứ không phải** $n-1$. PNG

Joint Estimation

Model: $X_i \sim \mathcal{N}(\mu, \sigma^2)$

MLEs: $\hat{\mu}_{MLE} = \overline{X}_n$, $\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(x_i - \overline{X}_n)^2$

Note: MLE uses n, not $n-1$

1. Xuất phát từ phép giải đại số (Toán học thuần túy)

Phương pháp MLE không tự quy định chọn mẫu số n hay $n-1$. Mẫu số n tự động xuất hiện khi giải phương trình đạo hàm tìm điểm cực đại của hàm Log-Likelihood.

Xét hàm Log-Likelihood theo tham số σ^2 (sau khi đã thay $\hat{\mu} = \bar{X}_n$):

$$\ell(\sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \bar{X}_n)^2$$

Đạo hàm riêng theo σ^2 và cho bằng 0:

$$\frac{d\ell}{d(\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i - \bar{X}_n)^2 = 0$$

Chuyển vế và triệt tiêu các hệ số:

$$\frac{n}{2\sigma^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{X}_n)^2}{2(\sigma^2)^2}$$

Nhân cả hai vế với $2(\sigma^2)^2$:

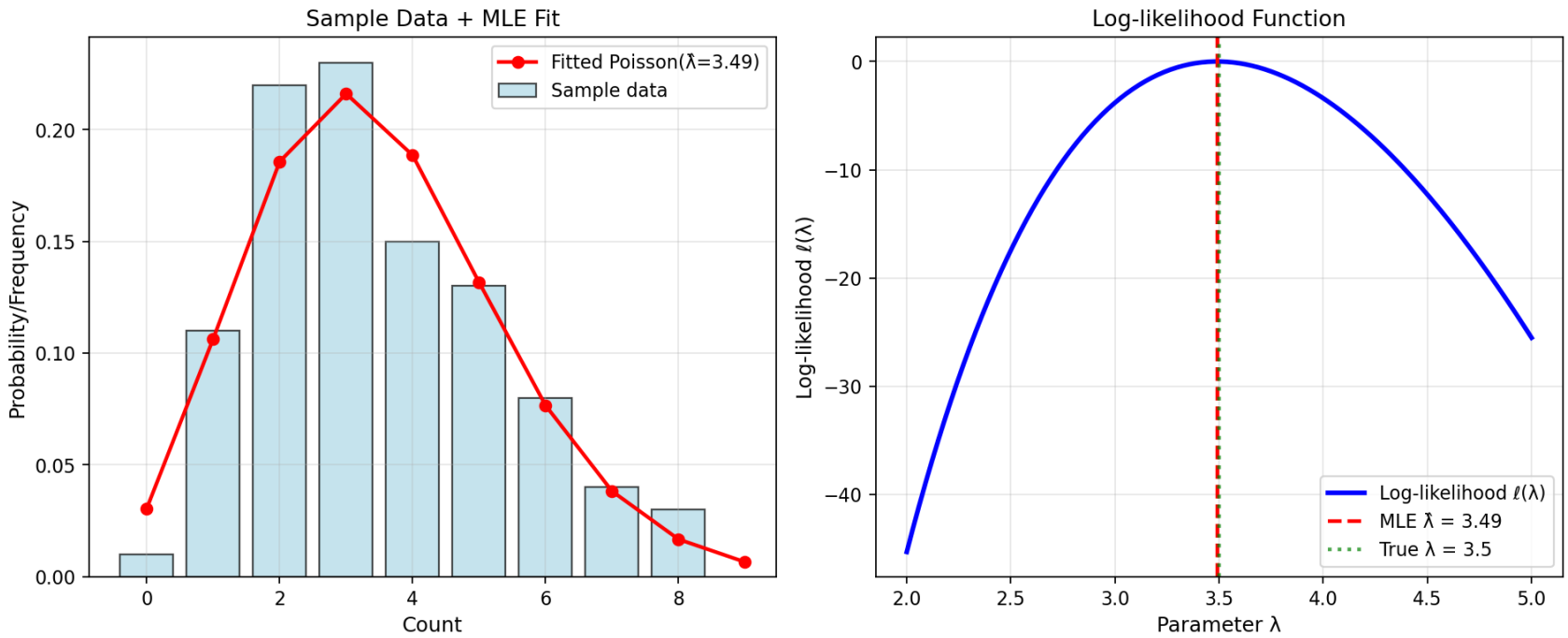
$$n \cdot \sigma^2 = \sum_{i=1}^{n}(x_i - \bar{X}_n)^2 \implies \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{X}_n)^2$$

Số n ở mẫu số xuất phát trực tiếp từ số hạng $-\frac{n}{2}\log(\sigma^2)$. Số hạng này có được do ta nhân tích của n hàm mật độ xác suất độc lập lại với nhau.

2. Ý nghĩa bản chất: Mục tiêu của MLE khác với Ước lượng Không Chệch

- **Mục tiêu của MLE:** Tìm tham số làm cho tập dữ liệu n điểm hiện có **đạt xác suất xuất hiện cao nhất**. Về mặt hình học, giá trị phương sai tạo ra đỉnh cao nhất cho hàm mật độ chính là trung bình bình phương khoảng cách của đúng n điểm dữ liệu đó quanh $\bar{X}_n$. Vì có n điểm dữ liệu, trung bình cộng bắt buộc phải chia cho n.
- **Mẫu số $n-1$ là do hiệu chỉnh bổ sung (Bessel's Correction):** Khi dùng trung bình mẫu $\bar{X}_n$ thay cho trung bình thực μ, tổng bình phương khoảng cách $\sum(x_i - \bar{X}_n)^2$ luôn nhỏ hơn tổng bình phương khoảng cách thực $\sum(x_i - \mu)^2$. Nếu chia cho n, giá trị ước lượng bị nhỏ hơn thực tế một chút (bị chệch). Để sửa độ chệch này, thống kê học cổ điển hiệu chỉnh mẫu số thành $n-1$ (mất 1 bậc tự do do phải ước lượng $\bar{X}_n$).

MLE for Poisson Parameter λ



Discrete Distribution MLE

Model: $X_i \sim \text{Poisson}(\lambda)$

Log-likelihood: $\ell(\lambda) = \sum(x_i \log \lambda - \lambda - \log(x_i!))$

MLE: $\hat{\lambda}_{MLE} = \overline{X}_n$

Theoretical Properties of MLE

1. Consistency: $\hat{\theta}_{MLE} \xrightarrow{P} \theta$
   - **Mathematical Definition:** As the sample size n approaches infinity, the estimator $\hat{\theta}_{MLE}$ converges in probability to the true parameter value θ_0. For any $\epsilon > 0$: PNG

$$\lim_{n \to \infty} P(|\hat{\theta}_{MLE} - \theta_0| < \epsilon) = 1$$

   - **Left Plot Interpretation:** The plot shows the probability density function of $\hat{\theta}$ across different sample sizes centered at true $\theta = 2.00$: PNG
     - At $n = 10$, the distribution is wide, meaning high sampling variability. PNG
     - At $n = 200$, the distribution collapses into a sharp spike around $\theta = 2.00$. As $n \to \infty$, all probability mass concentrates at the exact true parameter. PNG +1

2. Asymptotic Normality: $\sqrt{n}(\hat{\theta}_{MLE} - \theta) \xrightarrow{d} \mathcal{N}\left(0, I(\theta)^{-1}\right)$
   - **Mathematical Definition:** For large sample sizes n, the sampling distribution of $\hat{\theta}_{MLE}$ is approximately Gaussian: PNG

$$\hat{\theta}_{MLE} \sim \mathcal{N}\left(\theta, \frac{1}{nI(\theta)}\right)$$

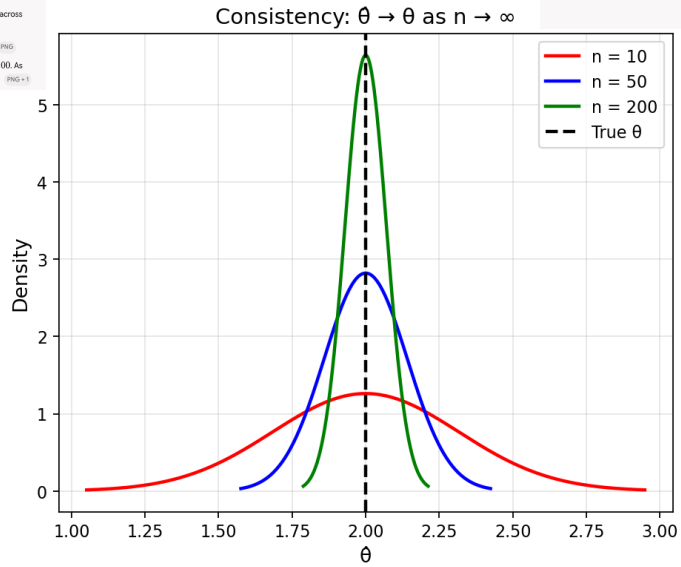
     where $I(\theta)$ is the **Fisher Information per observation.** PNG
   - **Key Implication:** Regardless of the underlying distribution of the individual data points X_i (whether Bernoulli, Poisson, Exponential, etc.), the distribution of the estimator $\hat{\theta}_{MLE}$ itself becomes Normal as $n \to \infty$ due to the Central Limit Theorem. This enables hypothesis testing (z-tests) and construction of asymptotic confidence intervals:

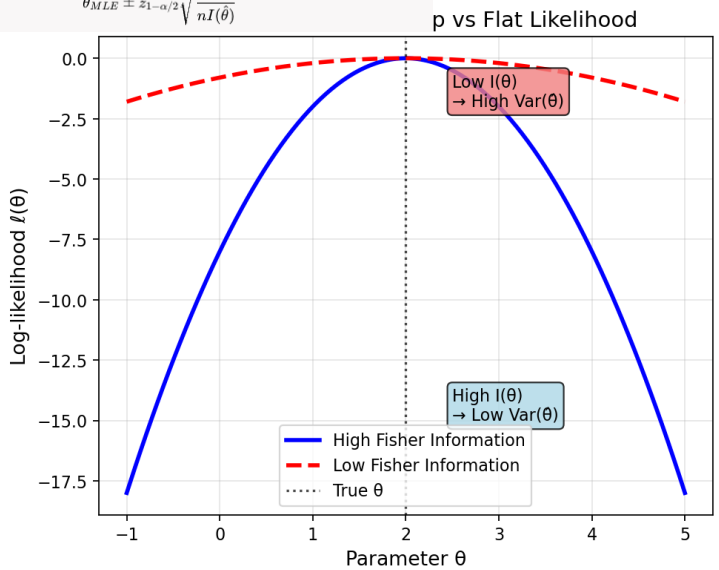
$$\hat{\theta}_{MLE} \pm z_{1-\alpha/2}\sqrt{\frac{1}{nI(\hat{\theta})}}$$

3. Asymptotic Efficiency & Fisher Information (Right Plot)
   - **Cramér-Rao Lower Bound (CRLB):** The Cramér-Rao inequality states that the variance of *any* unbiased estimator $T(X)$ is bounded below by the inverse of the total Fisher Information:

$$\text{Var}(T(X)) \geq \frac{1}{nI(\theta)}$$

   - **Efficiency:** MLE is **asymptotically efficient**, meaning as $n \to \infty$, its variance achieves this theoretical minimum bound. No unbiased estimator can achieve a lower variance than MLE in large samples. PNG





- **Consistency:** $\hat{\theta}_{MLE} \to \theta$
- **Asymptotic Normality:** $\sqrt{n}(\hat{\theta}_{MLE} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1})$
- **Efficiency:** achieves the CramérRao lower bound asymptotically

Step D: Applying Slutsky's Theorem

Dividing a sequence that converges in distribution to $\mathcal{N}(0, I(\theta_0))$ by a term converging in probability to constant $I(\theta_0)$ gives:

$$\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \frac{\mathcal{N}(0, I(\theta_0))}{I(\theta_0)} = \mathcal{N}\left(0, \frac{I(\theta_0)}{(I(\theta_0))^2}\right) = \mathcal{N}\left(0, I(\theta_0)^{-1}\right)$$

2. Why MLE Becomes Normally Distributed (Proof Outline)

The emergence of the Gaussian curve is a direct consequence of applying the **Central Limit Theorem (CLT)** and the **Law of Large Numbers (LLN)** to the log-likelihood function.

Step A: First-Order Taylor Expansion

Since $\hat{\theta}_{\text{MLE}}$ maximizes the log-likelihood $\ell(\theta)$, the score function (first derivative) evaluated at $\hat{\theta}_{\text{MLE}}$ equals zero:

$$\ell'(\hat{\theta}_{\text{MLE}}) = 0$$

Expanding $\ell'(\hat{\theta}_{\text{MLE}})$ in a Taylor series around the true parameter θ_0:

$$0 = \ell'(\hat{\theta}_{\text{MLE}}) \approx \ell'(\theta_0) + \ell''(\theta_0)(\hat{\theta}_{\text{MLE}} - \theta_0)$$

Rearranging terms yields:

$$\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \approx \frac{\frac{1}{\sqrt{n}}\ell'(\theta_0)}{-\frac{1}{n}\ell''(\theta_0)}$$

Step B: Numerator Behavior via CLT

The numerator is the scaled score function, which is a sum of independent random variables: $\ell'(\theta_0) = \sum_{i=1}^{n} \frac{\partial \log f(x_i|\theta_0)}{\partial \theta}$.

- Expected value of score: $\mathbb{E}[\ell'(\theta_0)] = 0$.
- Variance of score: $\text{Var}(\ell'(\theta_0)) = nI(\theta_0)$.

By the Central Limit Theorem:

$$\frac{1}{\sqrt{n}}\ell'(\theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0))$$

Step C: Denominator Behavior via LLN

By the Weak Law of Large Numbers, the average of second derivatives converges in probability to its expected value (the Fisher Information):

$$-\frac{1}{n}\ell''(\theta_0) \xrightarrow{p} I(\theta_0)$$

3. Practical Significance in Statistical Inference

Without asymptotic normality, constructing exact confidence intervals or hypothesis tests for complex non-Gaussian models would require deriving difficult, model-specific finite-sample distributions analytically.

Asymptotic normality provides universal tools for large samples:

1. **Standard Errors:**

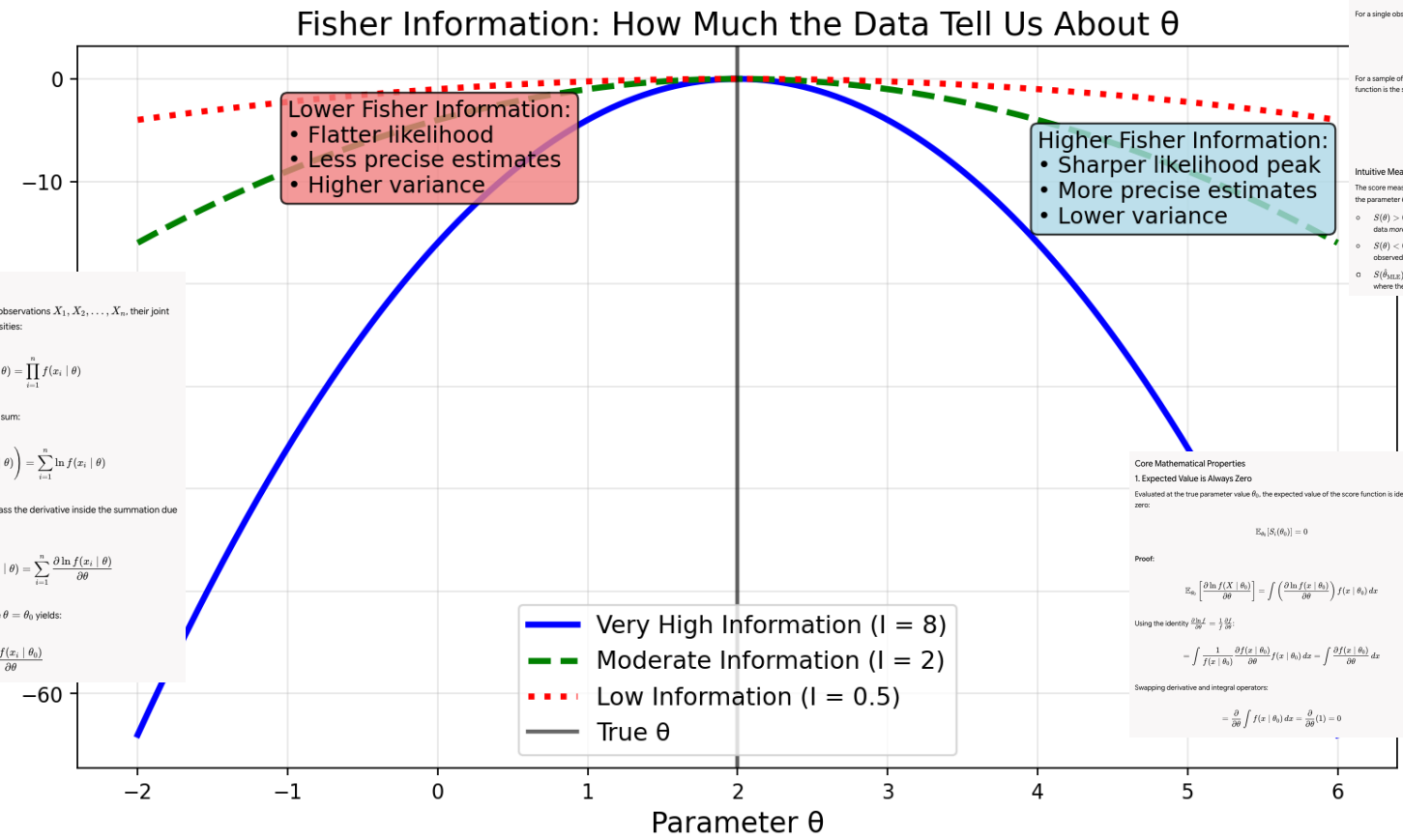
$$\text{SE}(\hat{\theta}) = \frac{1}{\sqrt{nI(\hat{\theta})}}$$

2. **Wald Confidence Intervals:**

$$\hat{\theta}_{\text{MLE}} \pm z_{1-\alpha/2} \cdot \text{SE}(\hat{\theta})$$

3. **Wald Hypothesis Testing:**
   To test $H_0 : \theta = \theta_0$, the statistic $W = \frac{\hat{\theta}_{\text{MLE}} - \theta_0}{\text{SE}(\hat{\theta})}$ follows a standard normal distribution $\mathcal{N}(0, 1)$ under H_0.

Fisher Information



1. The Algebraic Derivation

For n independent and identically distributed (i.i.d.) observations $X_1, X_2, \ldots, X_n$, their joint likelihood $L(\theta)$ is the product of their individual densities:

$$L(\theta) = f(x_1, x_2, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

Taking the natural logarithm turns the product into a sum:

$$\ell(\theta) = \ln L(\theta) = \ln\left(\prod_{i=1}^{n} f(x_i \mid \theta)\right) = \sum_{i=1}^{n} \ln f(x_i \mid \theta)$$

Taking the derivative with respect to θ allows us to pass the derivative inside the summation due to the linearity of differentiation:

$$\ell'(\theta) = \frac{\partial}{\partial\theta}\ell(\theta) = \frac{\partial}{\partial\theta}\sum_{i=1}^{n} \ln f(x_i \mid \theta) = \sum_{i=1}^{n} \frac{\partial \ln f(x_i \mid \theta)}{\partial\theta}$$

Evaluating this derivative at the true parameter value $\theta = \theta_0$ yields:

$$\ell'(\theta_0) = \sum_{i=1}^{n} \frac{\partial \ln f(x_i \mid \theta_0)}{\partial\theta}$$

The **Score function** $S(\theta)$ (or simply **the Score**) is the first derivative (gradient) of the log-likelihood function with respect to the parameter θ.

For a single observation X_i, the individual score variable $S_i(\theta)$ is:

$$S_i(\theta) = \frac{\partial}{\partial\theta} \ln f(X_i \mid \theta)$$

For a sample of n independent and identically distributed (i.i.d.) observations, the total score function is the sum of the individual scores:

$$S(\theta) = \frac{\partial}{\partial\theta}\ell(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial\theta} \ln f(X_i \mid \theta)$$

Intuitive Meaning

The score measures how **sensitive** the log-likelihood of the observed data is to small changes in the parameter θ:

- $S(\theta) > 0$: The log-likelihood curve is sloping upward. Increasing θ makes the observed data *more* likely.
- $S(\theta) < 0$: The log-likelihood curve is sloping downward. Increasing θ makes the observed data less likely.
- $S(\hat{\theta}_{MLE}) = 0$: At the Maximum Likelihood Estimator $\hat{\theta}_{MLE}$, the curve reaches its peak, where the score drops to zero (the first-order optimality condition).

Core Mathematical Properties

1. Expected Value is Always Zero

Evaluated at the true parameter value θ_0, the expected value of the score function is identically zero:

$$\mathbb{E}_{\theta_0}[S_i(\theta_0)] = 0$$

Proof:

$$\mathbb{E}_{\theta_0}\left[\frac{\partial \ln f(X \mid \theta_0)}{\partial\theta}\right] = \int \left(\frac{\partial \ln f(x \mid \theta_0)}{\partial\theta}\right) f(x \mid \theta_0)\, dx$$

Using the identity $\frac{\partial \ln f}{\partial\theta} = \frac{1}{f}\frac{\partial f}{\partial\theta}$:

$$= \int \frac{1}{f(x \mid \theta_0)} \frac{\partial f(x \mid \theta_0)}{\partial\theta} f(x \mid \theta_0)\, dx = \int \frac{\partial f(x \mid \theta_0)}{\partial\theta}\, dx$$

Swapping derivative and integral operators:

$$= \frac{\partial}{\partial\theta}\int f(x \mid \theta_0)\, dx = \frac{\partial}{\partial\theta}(1) = 0$$

Leibniz Integral Rule

The general rule for differentiating under the integral sign (Leibniz's Rule) is:

$$\frac{d}{d\theta}\int_{a(\theta)}^{b(\theta)} f(x,\theta)\, dx = \underbrace{f(b(\theta),\theta)\cdot b'(\theta) - f(a(\theta),\theta)\cdot a'(\theta)}_{\text{Boundary terms}} + \int_{a(\theta)}^{b(\theta)} \frac{\partial}{\partial\theta} f(x,\theta)\, dx$$

Notice those two extra terms at the front! We can only drop them and pull the derivative directly inside the integral if two core conditions hold:

1. **Fixed Support (Constant Limits):** The integration bounds $[a(\theta), b(\theta)]$ (the region where $f(x \mid \theta) > 0$) **do not depend on** θ.
2. **Smoothness:** $f(x \mid \theta)$ is sufficiently smooth and bounded so that the Dominated Convergence Theorem applies.

For distributions like Gaussian, Poisson, or Exponential, the limits are $(-\infty, \infty)$ or $[0, \infty)$, which don't change with θ. Thus, $b'(\theta) = 0$ and $a'(\theta) = 0$, making the boundary terms vanish completely!

Definition & Intuition

Fisher Information: $I(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\ell(\theta)\right]$

High information sharper peak lower variance

Fisher Information $I(\theta)$ measures how much information an observed sample carries about an unknown parameter θ 📊.

Here are two ways to understand it:

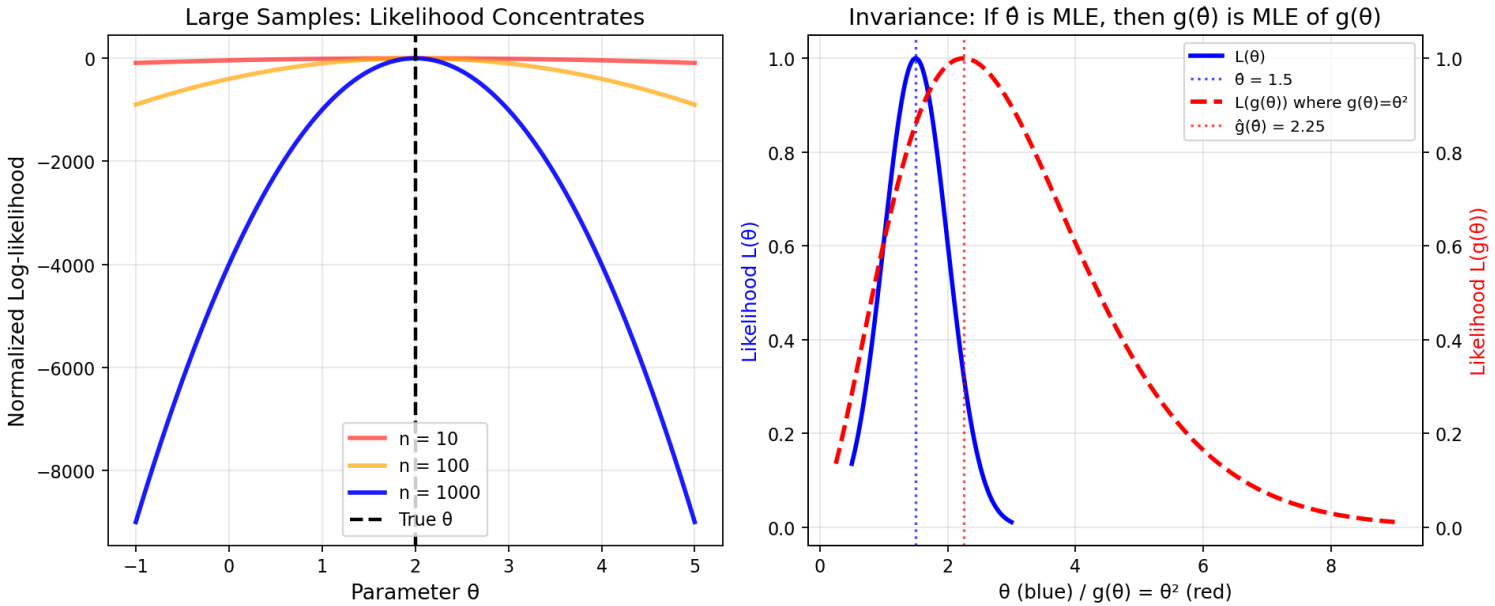
1. 🎯 **Intuitive View (Curvature):** Think of the shape of the log-likelihood curve.
   - **High Fisher Information:** The curve has a **sharp, steep peak** 📈. Small shifts away from the peak cause a rapid drop in likelihood, meaning the data gives us strong certainty about the true parameter.
   - **Low Fisher Information:** The curve is **flat** 📉. Many values of θ yield nearly identical likelihoods, meaning the data carries high uncertainty.
2. 📐 **Mathematical View (Curvature & Variance):** It is mathematically defined as the expected negative curvature (second derivative) of the log-likelihood function:

$$I(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \ln f(X \mid \theta)\right]$$

Equivalently, it is the variance of the score function $S(\theta) = \frac{\partial \ln f}{\partial\theta}$.

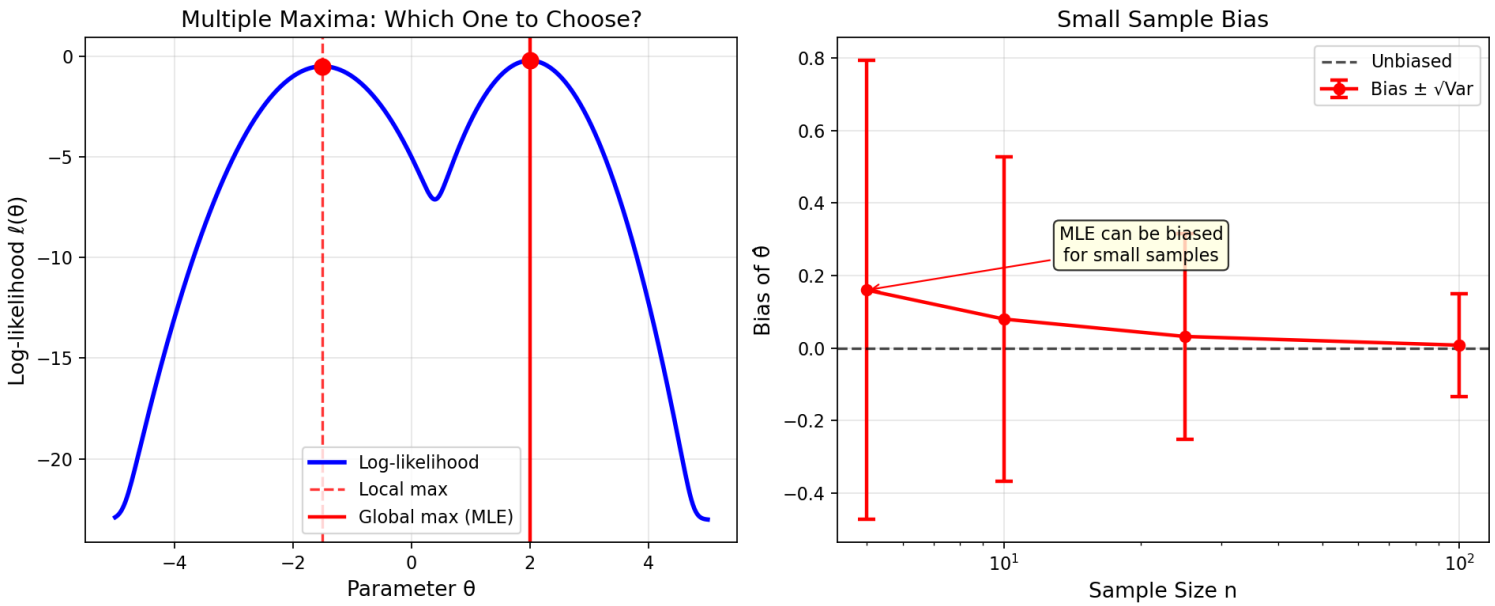
To connect this to estimation: If a dataset gives us a log-likelihood curve with a very **steep, sharp peak** (high Fisher Information), would you expect the variance of our estimate $\hat{\theta}$ to be **large** or **small**? 💡

Strengths of MLE



- Works well for large n (asymptotic guarantees)
- Very flexible, applicable to many models
- **Invariance property:** if $\hat{\theta}$ is MLE of θ, then $g(\hat{\theta})$ is MLE of $g(\theta)$

Limitations of MLE



- Small n bias and instability
- Non-identifiability multiple maxima
- Likelihood surface can be flat or multimodal
- Sensitive to model misspecification

MLE: Strengths and Limitations

✓ MLE Works Well When...	⚠ MLE May Struggle When...
✓ Sufficient data	⚠ Insufficient/poor data
✓ Regular likelihood surface	⚠ Multimodal likelihood
✓ Identifiable parameters	⚠ Non-identifiable parameters
✓ Correct model specification	⚠ Model misspecification
✓ Large sample size ($n \to \infty$)	⚠ Small sample size (finite n)

- MLE is powerful and widely used
- Asymptotically consistent, normal, and efficient
- Must be cautious with small samples or misspecified models

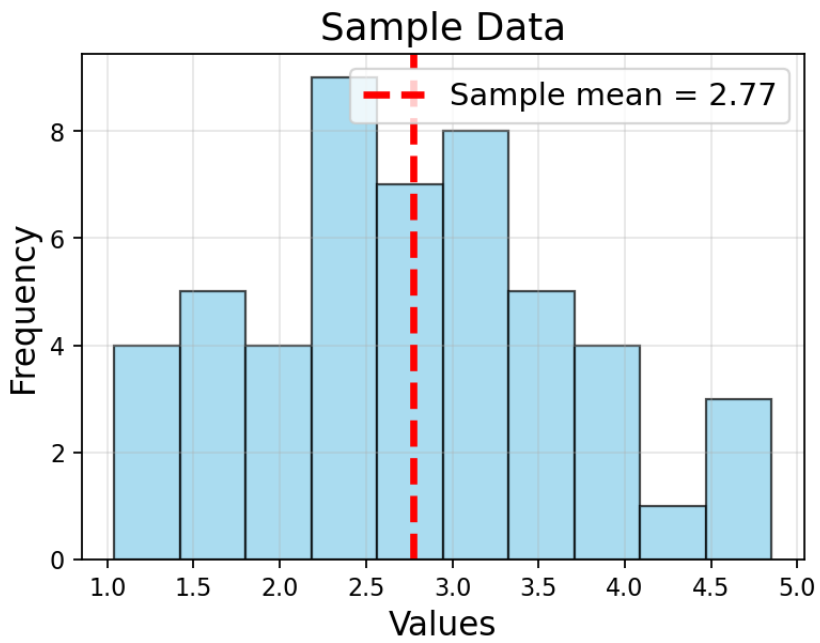
Outline

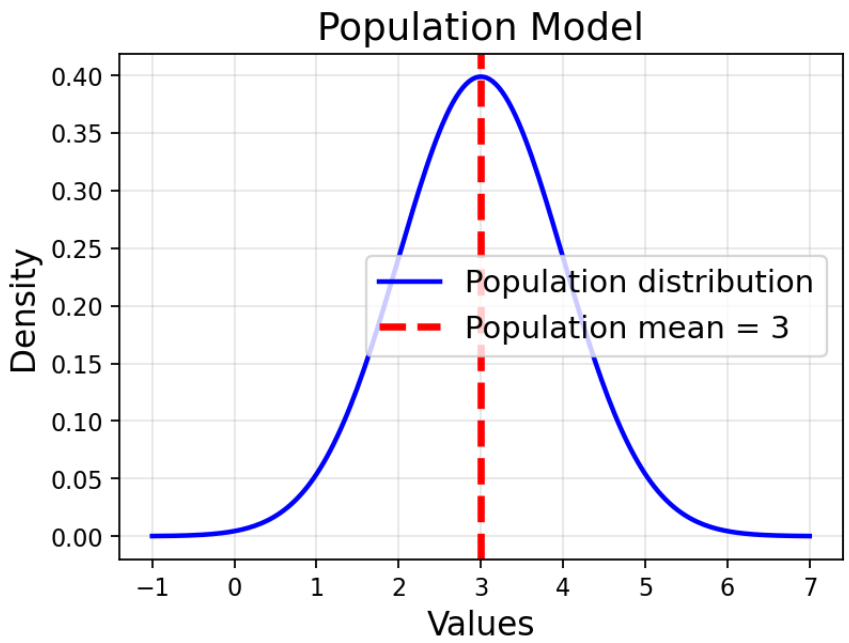
1 Maximum Likelihood Estimation (MLE)

2 Method of Moments

Why another estimation method?

- MLE is powerful, but sometimes hard to compute.
- Method of Moments (MoM) offers a simpler alternative.
- Idea: match sample moments with theoretical moments.





The Method of Moments

- For model parameter θ, theoretical moment:

$$m_k(\theta) = \mathbb{E}_\theta[X^k].$$

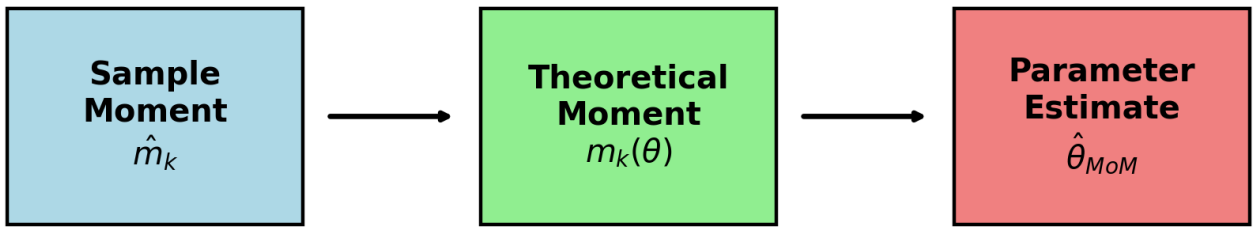
- Empirical moment:

$$\hat{m}_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

- Solve equations:

$$m_k(\theta) = \hat{m}_k.$$

Method of Moments Principle



$m_k(\theta) = \hat{m}_k$

MoM for Bernoulli: Step-by-Step Derivation

Setting up the system of equations:

$$\text{Theoretical first moment:} \quad E[X] = p \tag{1}$$

$$\text{Sample first moment:} \quad \hat{m}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}_n \tag{2}$$

Method of Moments equation:

$$E[X] = \hat{m}_1 \tag{3}$$

$$p = \overline{X}_n \tag{4}$$

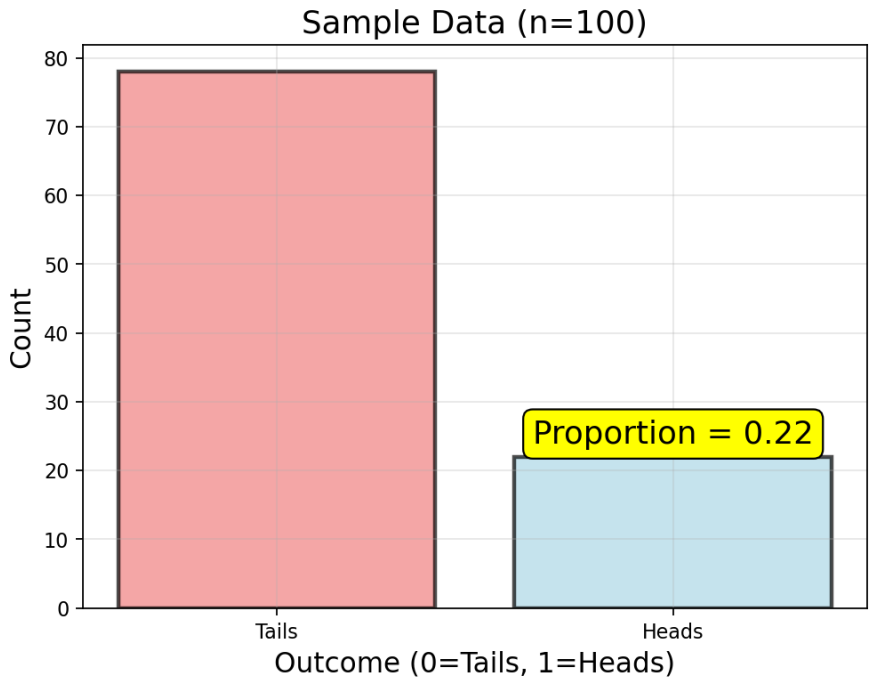
Solving for the parameter:

$$\hat{p}_{MoM} = \overline{X}_n \tag{5}$$

This shows that MoM reduces to solving a simple algebraic equation!

MoM for Bernoulli(p)

- Theoretical mean: $E[X] = p$.
- Sample mean: $\hat{m}_1 = \overline{X}_n$.
- Solve: $\hat{p}_{MoM} = \overline{X}_n$.
- Note: coincides with MLE.



Method of Moments:

$$E[X] = p$$

$$\hat{m}_1 = \bar{X}_n$$

$$\hat{p}_{MoM} = \bar{X}_n$$

$$\hat{p}_{MoM} = 0.22$$

MoM for Poisson: Step-by-Step Derivation

Setting up the system of equations:

$$\text{Theoretical first moment:} \quad E[X] = \lambda \tag{6}$$

$$\text{Sample first moment:} \quad \hat{m}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}_n \tag{7}$$

Method of Moments equation:

$$E[X] = \hat{m}_1 \tag{8}$$

$$\lambda = \overline{X}_n \tag{9}$$

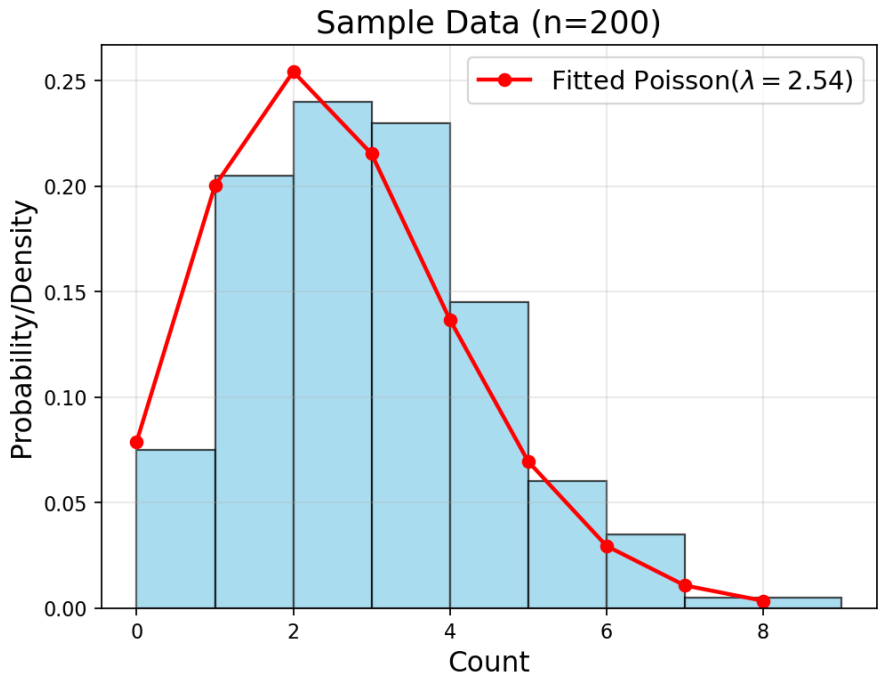
Solving for the parameter:

$$\hat{\lambda}_{MoM} = \overline{X}_n \tag{10}$$

Again, we solve a simple equation: parameter = sample mean!

MoM for Poisson(λ)

- Theoretical mean: $E[X] = \lambda$.
- Sample mean: $\hat{m}_1 = \overline{X}_n$.
- Solve: $\hat{\lambda}_{MoM} = \overline{X}_n$.
- Note: coincides with MLE.



Method of Moments:

$$E[X] = \lambda$$

$$\hat{m}_1 = \bar{X}_n$$

$$\hat{\lambda}_{MoM} = \bar{X}_n$$

$$\hat{\lambda}_{MoM} = 2.54$$

MoM for Normal: Step-by-Step Derivation

Setting up the system of equations (2 parameters $\Rightarrow$ 2 moments):

$$\text{Theoretical moments:} \quad E[X] = \mu, \quad E[X^2] = \mu^2 + \sigma^2 \tag{11}$$

$$\text{Sample moments:} \quad \hat{m}_1 = \overline{X}_n, \quad \hat{m}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 \tag{12}$$

Method of Moments equations:

$$\mu = \overline{X}_n \tag{13}$$

$$\mu^2 + \sigma^2 = \hat{m}_2 \tag{14}$$

Solving the system:

$$\hat{\mu}_{MoM} = \overline{X}_n \tag{15}$$

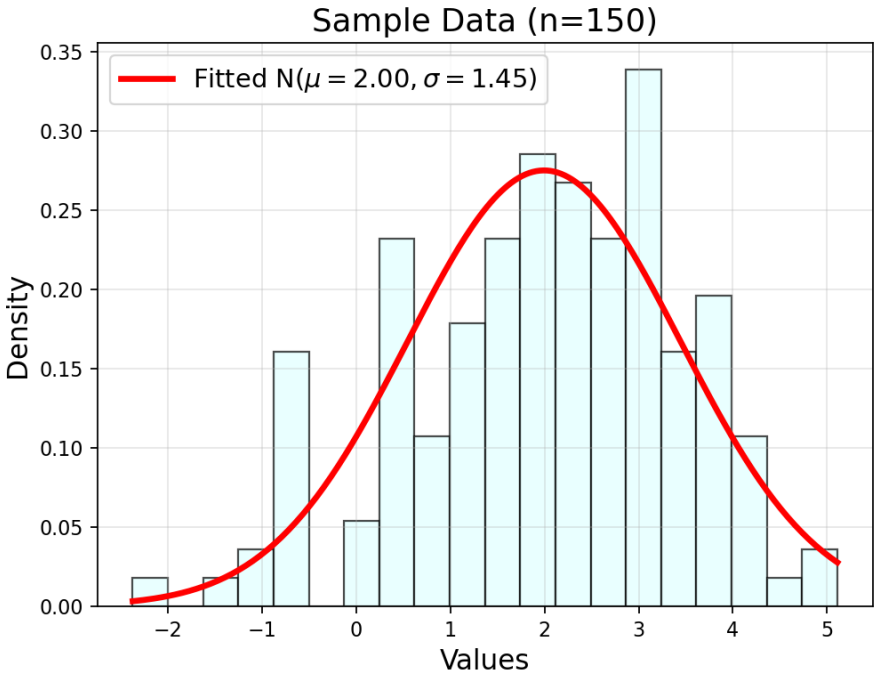
$$\hat{\sigma}^2_{MoM} = \hat{m}_2 - (\hat{\mu}_{MoM})^2 = \hat{m}_2 - (\overline{X}_n)^2 \tag{16}$$

Two parameters require solving a system of two equations!

MoM for Normal(μ, σ^2)

- $E[X] = \mu$, $E[X^2] = \mu^2 + \sigma^2$.
- Empirical moments: $\hat{m}_1 = \overline{X}_n$, $\hat{m}_2 = \frac{1}{n}\sum X_i^2$.
- Solve:

$$\hat{\mu}_{MoM} = \overline{X}_n, \quad \hat{\sigma}^2_{MoM} = \hat{m}_2 - (\overline{X}_n)^2.$$



Method of Moments:

$$E[X] = \mu, \quad E[X^2] = \mu^2 + \sigma^2$$

$$\hat{m}_1 = \bar{X}_n, \quad \hat{m}_2 = \frac{1}{n}\sum X_i^2$$

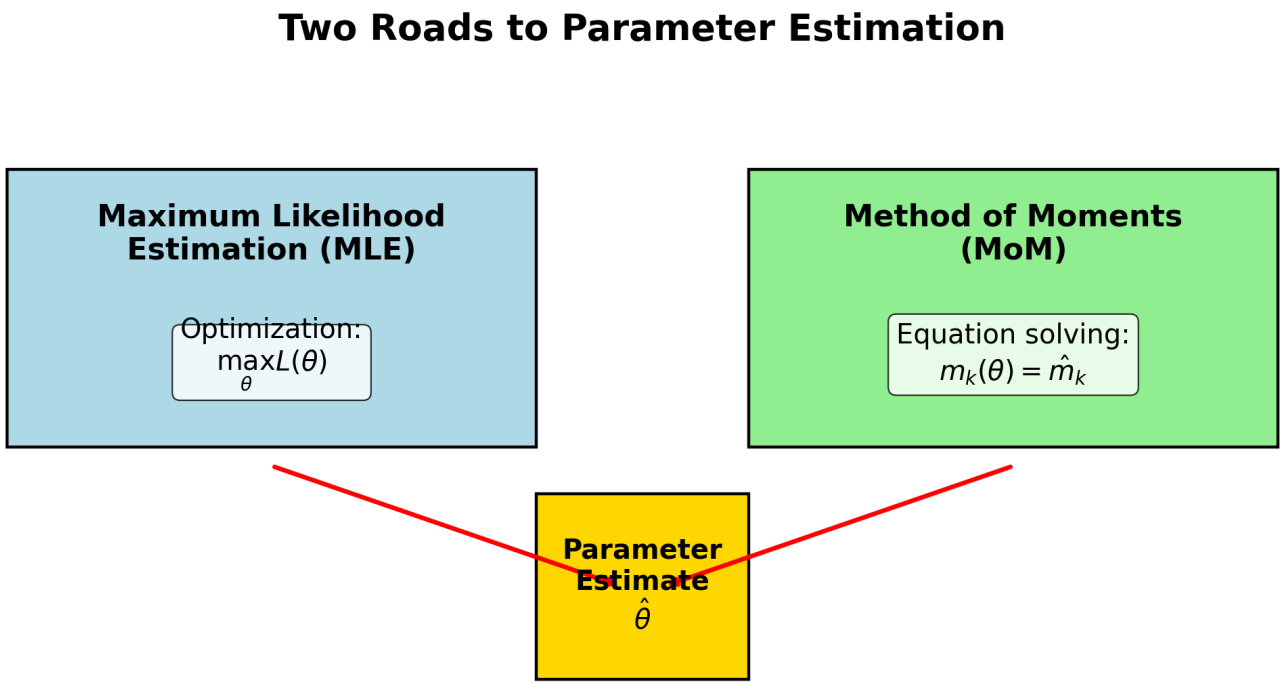
$$\hat{\mu}_{MoM} = \bar{X}_n$$

$$\hat{\sigma}^2_{MoM} = \hat{m}_2 - (\bar{X}_n)^2$$

$\hat{\mu}_{MoM} = 2.00, \hat{\sigma}_{MoM} = 1.45$

MLE vs MoM: Similarities and Differences

- Both provide consistent estimators (under conditions).
- MLE is asymptotically efficient; MoM is not guaranteed to be.
- MoM often easier to compute (simple equations).
- MoM can give nonsensical estimates (e.g., negative variance).
- In simple models, MLE = MoM.



Normal Variance Estimate (MLE vs MoM)

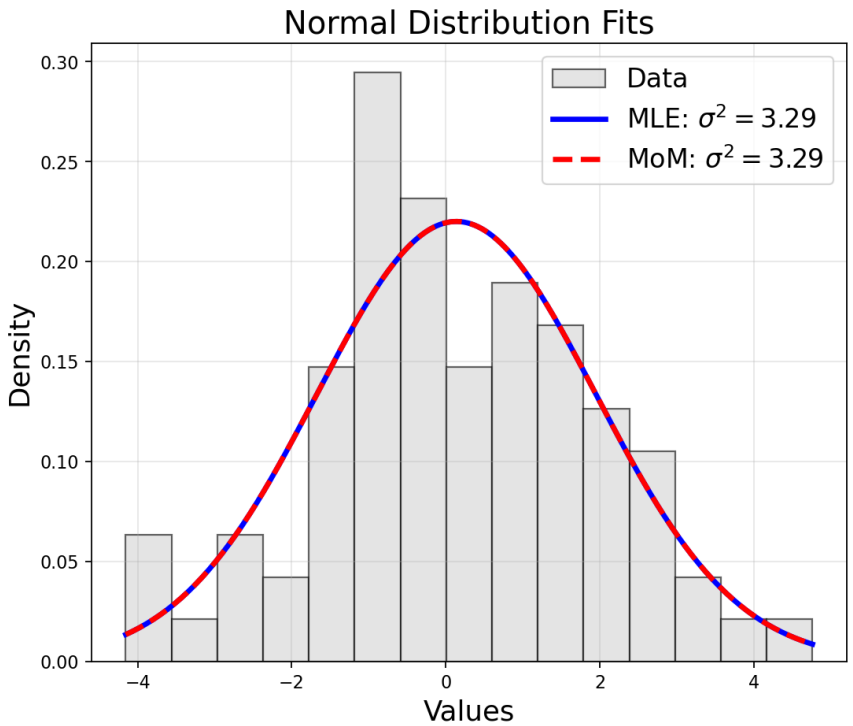
- MLE:

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(X_i - \overline{X}_n)^2.$$

- MoM:

$$\hat{\sigma}^2_{MoM} = \hat{m}_2 - (\overline{X}_n)^2.$$

- Here they are equal, but in other models they may differ.



Variance Estimators:

MLE:

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(X_i - \bar{X}_n)^2$$

MoM:

$$\hat{\sigma}^2_{MoM} = \hat{m}_2 - (\bar{X}_n)^2$$

In this case: MLE = 3.288, MoM = 3.288

Note: These are equal for Normal distribution!

MLE vs MoM: Summary Comparison

Property	MLE	MoM
Computation	Requires optimization	Simple equations
Large-sample behavior	Consistent, efficient	Consistent, less efficient
Small-sample behavior	Biased but systematic	Can be unstable, nonsensical
Flexibility	Works for many models	Requires finite moments
Coincidence	Often equals MoM in simple cases	Same as MLE sometimes

Key Takeaways: Method of Moments

- MoM: intuitive, simple, often first attempt.
- MLE: more powerful, statistically optimal in large samples.
- Both are fundamental tools in parameter estimation.

